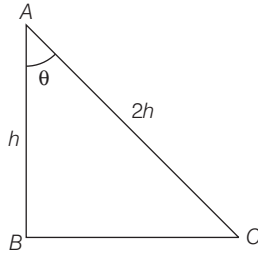


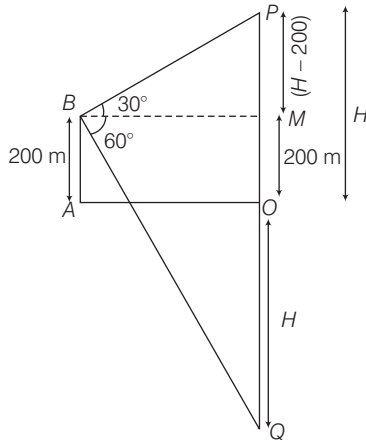
We know that, $\cos^2 \theta + \sin^2 \theta = 1$
 $\Rightarrow \left(\frac{BD}{300}\right)^2 + \left(\frac{BD}{200}\right)^2 = 1$
 [from Eqs. (i) and (ii)]
 $\Rightarrow BD^2 \left[\frac{40000 + 90000}{90000 \times 40000} \right] = 1$
 $\Rightarrow BD^2 \left[\frac{130000}{3600000000} \right] = 1$
 $\Rightarrow BD^2 = \frac{360000}{13}$
 $\therefore BD = \sqrt{\frac{360000}{13}} = \frac{600}{\sqrt{13}} \text{ m}$

39. (d) Let AB be the height of the post, AC be the string and the angle made by string with the post be θ .



Now, $\cos \theta = \frac{AB}{AC} = \frac{h}{2h} \Rightarrow \frac{1}{2} = \cos 60^\circ$
 $\therefore \theta = \frac{\pi}{3}$

40. (c) Let P be the cloud at a height H m above the level of the water in the lake and Q be its image in the water.

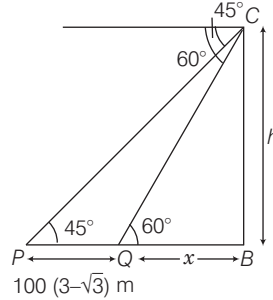


$\therefore OQ = OP = H$
 Given, $\angle PBM = 30^\circ$ and $\angle MBQ = 60^\circ$
 In right angled $\triangle PBM$,
 $\tan 30^\circ = \frac{PM}{BM} = \frac{H - 200}{BM}$
 $\Rightarrow \frac{1}{\sqrt{3}} = \frac{H - 200}{BM}$
 $\Rightarrow BM = \sqrt{3}(H - 200) \dots(i)$

In right angled $\triangle QMB$,
 $\tan 60^\circ = \frac{MQ}{BM} = \frac{H + 200}{BM}$
 $\Rightarrow \sqrt{3} = \frac{H + 200}{\sqrt{3}(H - 200)} \text{ [from Eq. (i)]}$
 $\Rightarrow H + 200 = 3(H - 200)$
 $\Rightarrow H + 200 = 3H - 600$
 $\Rightarrow 2H = 800$
 $\therefore H = 400 \text{ m}$

Hence, the height of the cloud is 400 m.

41. (c) Let $BC = h$ m be height of tower. Let P and Q be the points, where the angle subtended are 45° and 60° , respectively.

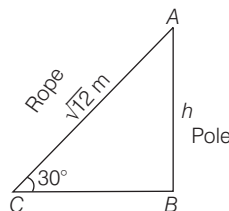


In right angled $\triangle BCQ$,
 $\tan 60^\circ = \frac{BC}{BQ} \Rightarrow \sqrt{3} = \frac{h}{x}$
 $\Rightarrow x = \frac{h}{\sqrt{3}} \dots(i)$

In right angled $\triangle PBC$,
 $\tan 45^\circ = \frac{BC}{PB} = \frac{BC}{PQ + QB}$
 $\Rightarrow 1 = \frac{h}{100(3 - \sqrt{3}) + x}$
 $\Rightarrow 100(3 - \sqrt{3}) + x = h$
 $\Rightarrow 100(3 - \sqrt{3}) + \frac{h}{\sqrt{3}} = h \text{ [from Eq. (i)]}$
 $\Rightarrow h - \frac{h}{\sqrt{3}} = 100(3 - \sqrt{3})$
 $\Rightarrow \frac{h(\sqrt{3} - 1)}{\sqrt{3}} = 100\sqrt{3}(\sqrt{3} - 1)$
 $\Rightarrow h = 100\sqrt{3} \times \sqrt{3}$
 $\therefore h = 300 \text{ m}$

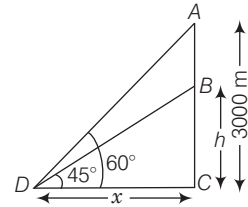
42. (d) Refer to example 3.

43. (d) AB is a pole and AC is rope.



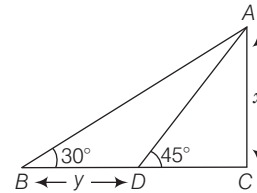
In $\triangle ABC$, $\sin 30^\circ = \frac{AB}{AC} = \frac{h}{\sqrt{12}}$
 $\Rightarrow \frac{h}{\sqrt{12}} = \frac{1}{2}$
 $\therefore h = \frac{\sqrt{12}}{2} = \frac{2\sqrt{3}}{2} = \sqrt{3} \text{ m}$

44. (c) Let A and B be the position of two planes and D be a point.



In $\triangle BCD$, $\tan 45^\circ = \frac{h}{x}$
 $\Rightarrow h = x$
 In $\triangle ACD$, $\tan 60^\circ = \sqrt{3} = \frac{3000}{x}$
 $\therefore x = \frac{3000}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 1000\sqrt{3} \text{ m}$
 $\therefore AB = 3000 - h$
 $= 3000 - 1000\sqrt{3}$
 $= 1000(3 - \sqrt{3}) \text{ m}$

45. (c) In right angled $\triangle ADC$, $\tan 45^\circ = \frac{AC}{DC}$
 $\Rightarrow DC = \frac{AC}{\tan 45^\circ} = x \dots(i)$



In right angled $\triangle BCA$, we get
 $\tan 30^\circ = \frac{AC}{BC} \Rightarrow \frac{x}{DC + BD}$
 $\Rightarrow \frac{1}{\sqrt{3}} = \frac{x}{y + x} \Rightarrow y + x = \sqrt{3}x$
 $\Rightarrow y = x(\sqrt{3} - 1)$
 $\Rightarrow \frac{x}{y} = \frac{1}{(\sqrt{3} - 1)}$
 $\Rightarrow \frac{x}{y} = \frac{1 \times (\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)}$
 $\Rightarrow \frac{x}{y} = \frac{(\sqrt{3} + 1)}{(\sqrt{3})^2 - (1)^2}$
 $\therefore \frac{x}{y} = \frac{(\sqrt{3} + 1)}{2}$

22

LINES AND ANGLES

Usually (2-3) questions have been asked from this chapter. And this is the easiest topic of geometry and so candidate can easily score marks in examinations.



In this chapter, we will study about the properties of the angles which formed, when two lines intersect each other and when a line intersect two or more parallel lines at distinct points.

Basic Terms and Definitions

Point

The figure of which length, breadth and height cannot be measured is called a point. It is infinitesimal.

Line Segment

The straight path between two points P and Q is called a line segment.

This can be represented as $\overline{PQ}$.

- P and Q are called the end points of the line segment.
- The line segment has a definite length.
- Distance between P and Q is called the length of the line segment PQ .

Line

A line segment extended endlessly on both sides, is

called a line. It is denoted by $\overleftrightarrow{PQ}$

or $\overleftrightarrow{QP}$.

- A line is a set of infinite number of points.
- A line has no end points and no definite length.

Ray

A ray extends indefinitely in one direction. This is exhibited by an arrow i.e. $\overrightarrow{PQ}$.

- P is called the initial point of the ray. The ray has no definite length.

Collinear Points

Three or more points are said to be collinear if a single straight line passes through them. Here, A , B and C are collinear.

Non-collinear Points

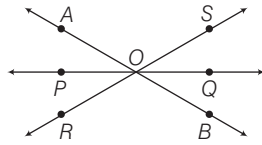
Three or more points not lying on a single straight line are called non-collinear points. Here, A , B and C are non-collinear points.

Intersecting Lines

Two lines having a common point are called intersecting lines. This common point is the point of intersection i.e. O .

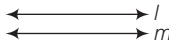
Concurrent Lines

Three or more lines intersecting at the same point are said to be concurrent. This common point is the point of concurrence i.e. O .



Non Intersecting lines/Parallel Lines

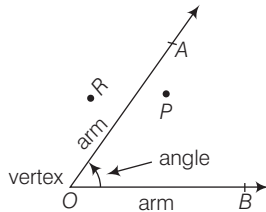
If two lines lie in the same plane and do not intersect when produced on either side, then such lines are said to be parallel to each other.



If l and m are two parallel lines, we write $l \parallel m$ and read it as l is parallel to m .

ANGLES

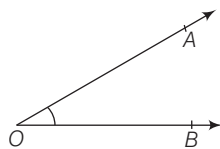
An angle is formed by two rays with a common initial point. Let 'O' is the initial point, then O is called the vertex.



Where, P is a point in the interior of $\angle AOB$ and R is a point in the exterior of $\angle AOB$.

Types of Angle

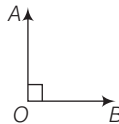
1. **Acute angle** An angle whose measure is more than 0° , but less than 90° , is called an acute angle.



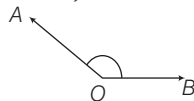
In the given figure, $0^\circ < \angle AOB < 90^\circ$

2. **Right angle** An angle whose measure is 90° , is called a right angle.

In the given figure, $\angle AOB = 90^\circ$.



3. **Obtuse angle** An angle whose measure is more than 90° but less than 180° , is called an obtuse angle.



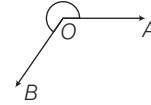
In the given figure, $90^\circ < \angle AOB < 180^\circ$.

4. **Straight angle** An angle whose measure is 180° , is called a straight angle.

In the figure, $\angle AOB = 180^\circ$



5. **Reflex angle** An angle whose measure is more than 180° , but less than 360° , is called a reflex angle.



In the given figure, $180^\circ < \angle AOB < 360^\circ$

6. **Complete angle** An angle whose measure is 360° , is called a complete angle. In the given figure, $\angle AOA$ is a complete angle.

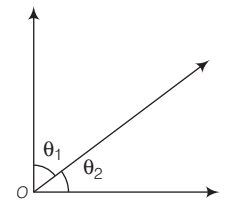


Pairs of Angle

1. Complementary Angles

Two angles are said to be complementary, if the sum of their measure is 90° . Thus, $\angle \theta_1$ and $\angle \theta_2$ are complementary, if $\angle \theta_1 + \angle \theta_2 = 90^\circ$

- Complementary angles are complement of each other.
- Complement of x is $(90^\circ - x)$.



EXAMPLE 1. The measure of an angle which is 28° more than its complement is

- a. 23° b. 59° c. 77° d. None of these

Sol. b. Let the measure of the required angle be x° . Then, measure of its complement $= 90^\circ - x$

$$\therefore x - (90^\circ - x) = 28^\circ \Leftrightarrow 2x = 118^\circ$$

$$\Leftrightarrow x = 59^\circ$$

Hence, the measure of the required angle is 59° .

EXAMPLE 2. The measure of the complement of an angle of $48^\circ 36' 24''$ is

- a. $41^\circ 23' 36''$ b. $42^\circ 23' 36''$ c. $41^\circ 24' 36''$ d. $42^\circ 24' 36''$

Sol. a. As, $90^\circ = 89^\circ 59' 60''$

$$\therefore \text{Complement of an angle of } (48^\circ 36' 24'')$$

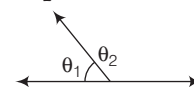
$$= \text{Angle of } [90^\circ - 48^\circ 36' 24'']$$

$$= \text{Angle of } [89^\circ 59' 60'' - 48^\circ 36' 24'']$$

$$= \text{Angle of } (41^\circ 23' 36'')$$

2. Supplementary Angles

Two angles are said to be supplementary, if the sum of their measures is 180° . Thus, $\angle \theta_1$ and $\angle \theta_2$ are supplementary if $\theta_1 + \theta_2 = 180^\circ$



- Supplementary angles are supplement of each other.
- Supplement of x is $(180^\circ - x)$.

EXAMPLE 3. The measure of an angle, which is 32° less than its supplement is

- a. 31° b. 64° c. 74° d. 148°

Sol. c. Let the measure of the required angle be x . Then, measure of its supplement $= (180^\circ - x)$

$$\therefore (180^\circ - x) - x = 32^\circ \Leftrightarrow 2x = 148^\circ \Leftrightarrow x = 74^\circ$$

EXAMPLE 4. Two supplementary angles are in the ratio 3 : 2. Then, the measurement of the smaller angle is

- a. 36° b. 72° c. 108° d. 112°

Sol. b. Let the supplementary angles be $3x$ and $2x$, respectively. Then, according to the definition of supplementary angle.

$$3x + 2x = 180^\circ$$

$$\Rightarrow 5x = 180^\circ \Rightarrow x = 36^\circ$$

$$\therefore \text{Angles will be } 3x = 3 \times 36 = 108^\circ$$

$$\text{and } 2x = 2 \times 36 = 72^\circ.$$

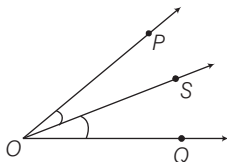
Thus, the smaller angle is 72° .

Bisector of an Angle

A ray which divides the angle into two equal parts, is called the bisector of an angle.

If a ray OS is the bisector of $\angle POQ$, then $\angle POS = \angle QOS$.

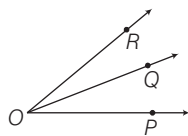
$$\bullet \angle POS = \angle QOS = \frac{1}{2} \angle POQ$$



Adjacent Angles

Two angles are said to be adjacent, if

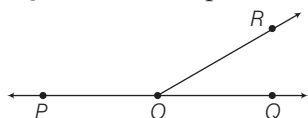
- they have a common vertex.
- they have a common arm
- their non-common arms are on either side of the common arm.



Here, $\angle POQ$ and $\angle ROQ$ are adjacent angles.

Linear Pairs of Angles

If the non-common arms of two adjacent angles form a line, then these angles are called linear pair of angles. $\angle ROP$ and $\angle ROQ$ form a linear pair of angles.

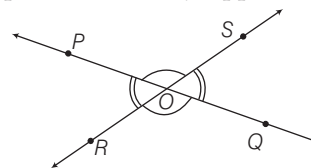


$$\therefore \angle POQ = \angle ROP + \angle ROQ = 180^\circ$$

Vertically Opposite Angles

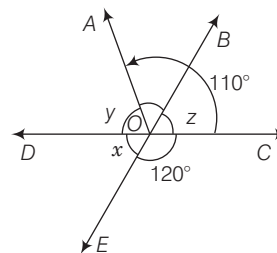
Two angles are called a pair of vertically opposite angles, if their arms form two pairs of opposite rays.

If two lines PQ and RS intersect at a point O , then the pair of $\angle POR$ and $\angle QOS$ or pair of $\angle POS$ and $\angle ROQ$ is said to be a pair of vertically opposite angles.



- Vertically opposite angles are always equal i.e. $\angle POS = \angle ROQ$ and $\angle POR = \angle SOQ$.
- Sum of the angles around a point is 360° .

EXAMPLE 5. In the given figure if DOC is a straight ray OB is bisector of $\angle AOC$, where $\angle AOC = 110^\circ$ and $\angle COE = 120^\circ$, then the sum of $\angle x$, $\angle y$ and $\angle z$ is



- a. 160° b. 115° c. 170° d. 180°

Sol. d. Since, DOC is a straight line

$$\angle AOC + \angle DOA = 180^\circ \quad [\text{Linear pair}]$$

$$110^\circ + \angle DOA = 180^\circ \quad [\because \angle AOC = 110^\circ]$$

$$\angle DOA = 70^\circ$$

$$\angle DOA = \angle y = 70^\circ$$

Also, ray OB is angle bisector of $\angle AOC$

$$\angle z = \frac{\angle AOC}{2} = \frac{110^\circ}{2} = 55^\circ$$

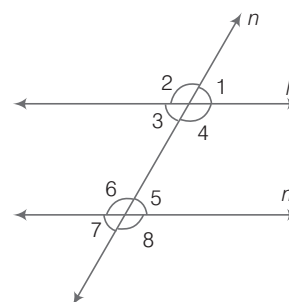
$\therefore$ Line BE and DC intersect each other at O .

$$\angle x = \angle z = 55^\circ \quad [\because \text{vertically opposite angle}]$$

$$\therefore \angle x + \angle y + \angle z = 55^\circ + 70^\circ + 55^\circ = 180^\circ$$

Some Other Angles

Let l and m are parallel lines and n is the transversal which cuts these parallel lines. The different angles formed are as follows.



- Corresponding angles** Corresponding angle pairs are
 $\angle 1$ and $\angle 5$, $\angle 2$ and $\angle 6$, $\angle 4$ and $\angle 8$, $\angle 3$ and $\angle 7$ and all the corresponding pair are equal
i.e., $\angle 1 = \angle 5$, $\angle 2 = \angle 6$, $\angle 4 = \angle 8$ and $\angle 3 = \angle 7$.
- Vertically opposite angles** Vertically opposite angles pairs are
 $\angle 1$ and $\angle 3$, $\angle 4$ and $\angle 2$, $\angle 8$ and $\angle 6$, $\angle 5$ and $\angle 7$ and all vertically opposite angles are equal
i.e., $\angle 1 = \angle 3$, $\angle 4 = \angle 2$, $\angle 8 = \angle 6$ and $\angle 5 = \angle 7$.
- Alternate angles** Alternate angles pairs are
 $\angle 3$ and $\angle 5$, $\angle 4$ and $\angle 6$ and they are equal
i.e., $\angle 3 = \angle 5$ and $\angle 4 = \angle 6$.
 - The sum of interior angles on the same side of transversal is equal to 180° .
i.e., $\angle 3 + \angle 6 = 180^\circ$ and $\angle 4 + \angle 5 = 180^\circ$
 - The sum of exterior angles on the same side of transversal is equal to 180° .
i.e., $\angle 2 + \angle 7 = 180^\circ$ and $\angle 1 + \angle 8 = 180^\circ$

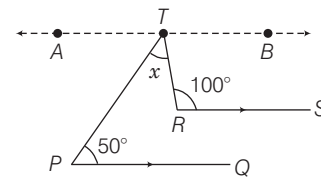
EXAMPLE 6. In the given figure, $PQ \parallel RS$. The value of x is

- a. 50° b. 80° c. 75° d. 65°

Sol. a. Draw $AB \parallel PQ$

$\therefore \angle ATP = \angle TPQ = 50^\circ$ [alternate angles]
 and $\angle BTR + \angle TRS = 180^\circ$

[Angles on the same side of transversal]



$\Rightarrow \angle BTR = 80^\circ$
 and $\angle ATP + x + \angle BTR = 180^\circ$ [$\because ATB$ is a straight line]
 $\therefore x = 180^\circ - (50^\circ + 80^\circ) = 50^\circ$

PRACTICE EXERCISE

- Three lines intersect each other in pairs. What is the total number of angles so formed?
 (a) 3 (b) 6 (c) 9 (d) 12
- An angle is 14° more than its complement. Then, its measure is
 (a) 166° (b) 86° (c) 76° (d) 52°
- The measure of an angle is twice the measure of its supplementary angle. So, its measure is
 (a) 120° (b) 60° (c) 100° (d) 90°
- What is the least number of straight lines for a bounded plane figure?
 (a) 1 (b) 2 (c) 3 (d) 4
- The supplement of $\frac{3}{5}$ of a right angle is
 (a) 122° (b) 126° (c) 130° (d) 132°
- In the given figure, if $PQ \parallel SR$, then the relation between $\angle a$ and $\angle b$ is

 (a) $\angle a \neq \angle b$ (b) $\angle a < \angle b$ (c) $\angle a = \angle b$ (d) $\angle a > \angle b$
- If $\angle 1 = (5x - 20)^\circ$ and $\angle 7 = (2x + 10)^\circ$, then $\angle 7$ is

 (a) 38° (b) 10°
 (c) 30° (d) 70°
- The measure of complementary angle of $12^\circ 25' 40''$ is
 (a) $77^\circ 34' 20''$ (b) $77^\circ 36' 20''$ (c) $77^\circ 24' 20''$ (d) $77^\circ 34'$
- $\angle POR$ and $\angle QOR$ form a linear pair and if $a - b = 80^\circ$, then the values of a and b are, respectively

 (a) $95^\circ, 85^\circ$ (b) $108^\circ, 72^\circ$
 (c) $130^\circ, 50^\circ$ (d) $105^\circ, 75^\circ$